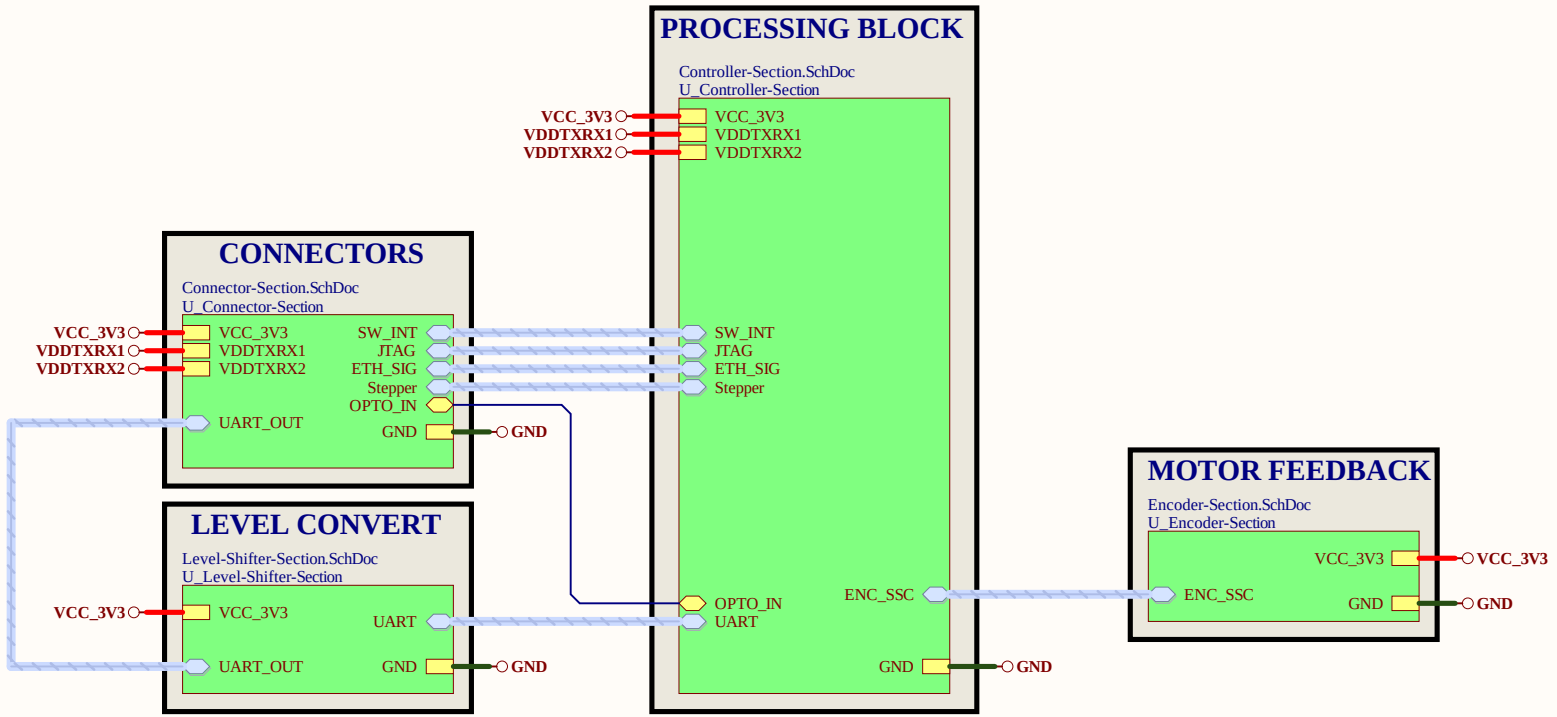
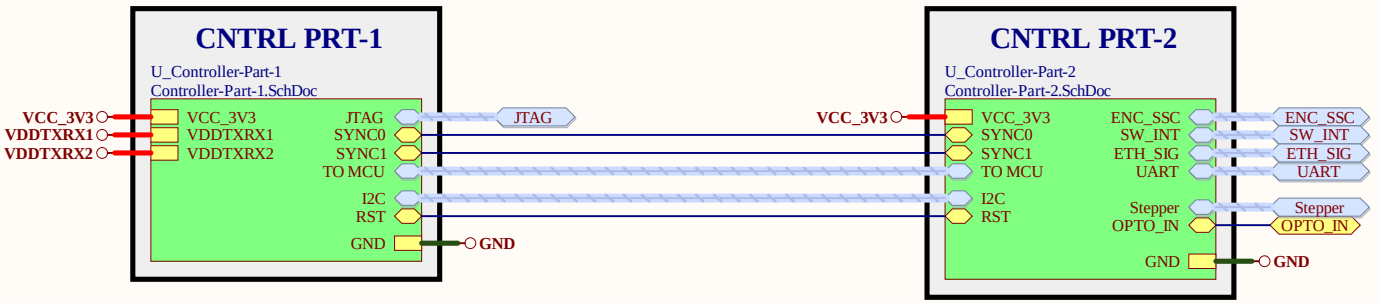
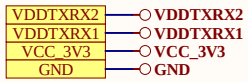
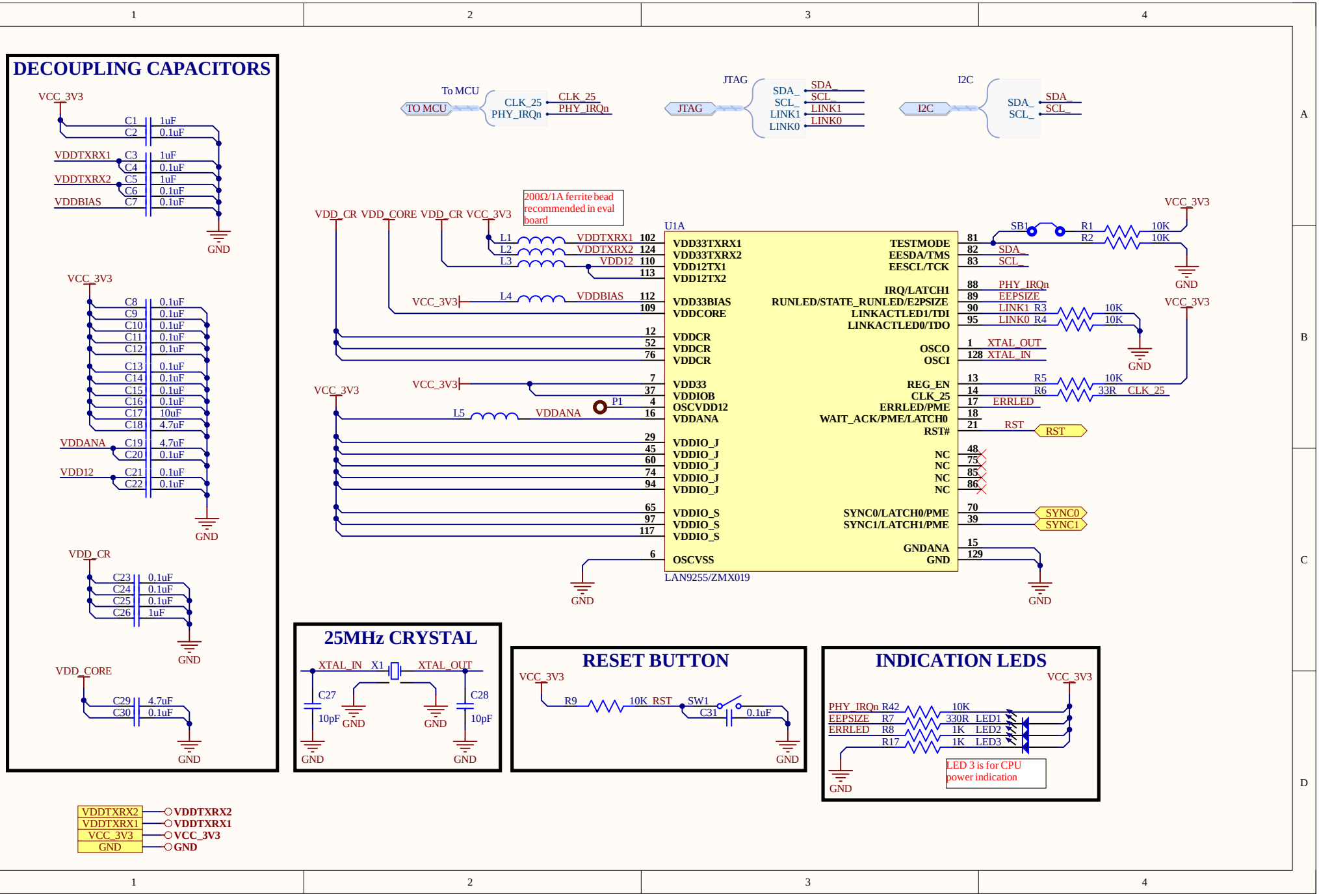
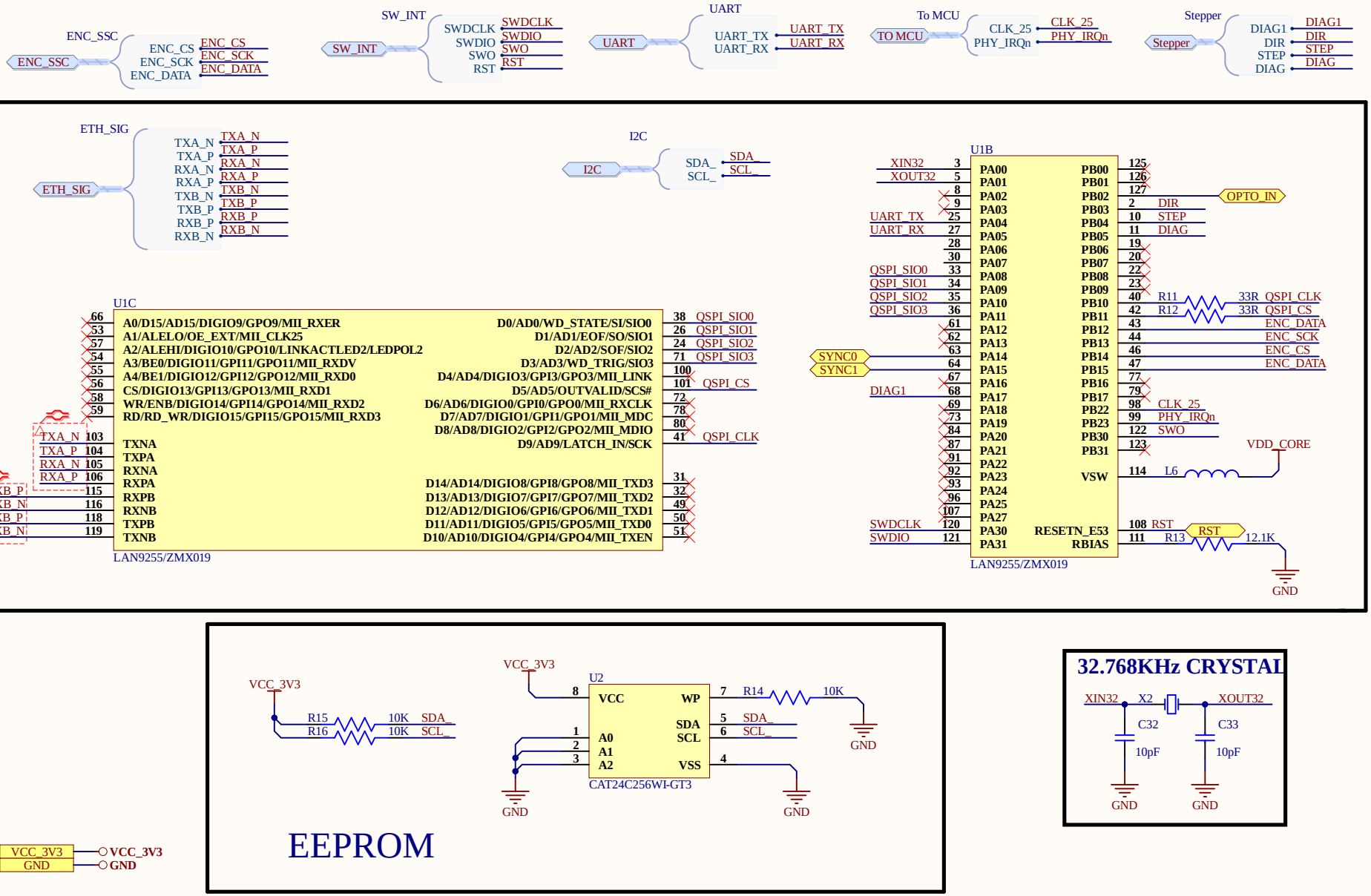

PROCESSING BLOCK
Controller-Section.SchDoc
U_Controller-Section
VCC_3V3
VDDTXRX1
VDDTXRX2
SW_INT
JTAG
ETH_SIG
Stepper
OPTO_IN
UART
ENC_SSC
GND
CONNECTORS
Connector-Section.SchDoc
U_Connector-Section
UART_OUT
LEVEL CONVERT
Level-Shifter-Section.SchDoc
U_Level-Shifter-Section
MOTOR FEEDBACK
Encoder-Section.SchDoc
U_Encoder-Section

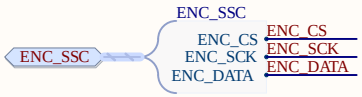

CNTRL PRT-1
U_Controller-Part-1
Controller-Part-1.SchDoc
VCC_3V3
VDDTXRX1
VDDTXRX2
JTAG
SYNC0
SYNC1
TO MCU
I2C
RST
GND
JTAG
GND
CNTRL PRT-2
U_Controller-Part-2
Controller-Part-2.SchDoc
VCC_3V3
SYNC0
SYNC1
TO MCU
I2C
RST
GND
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN
GND


VDDTXRX2
VDDTXRX1
VCC_3V3
GND
VDDTXRX2
VDDTXRX1
VCC_3V3
GND

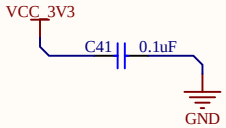

DECOUPLING CAPACITORS
VCC_3V3
C1 1uF
C2 0.1uF
VDDTXRX1 C3 1uF
C4 0.1uF
VDDTXRX2 C5 1uF
C6 0.1uF
VDDBIAS C7 0.1uF
GND
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
C12 0.1uF
C13 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 10uF
C18 4.7uF
VDDANA C19 4.7uF
C20 0.1uF
VDD12 C21 0.1uF
C22 0.1uF
VDD_CR
C23 0.1uF
C24 0.1uF
C25 0.1uF
C26 1uF
VDD_CORE
C29 4.7uF
C30 0.1uF
To MCU
TO MCU
CLK_25
PHY_IRQn
JTAG
SDA_
SCL_
LINK1
LINK0
I2C
200Ω/1A ferrite bead recommended in eval board
U1A
VDD33TXRX1 102
VDD33TXRX2 124
VDD12TX1 110
VDD12TX2 113
VDD33BIAS 112
VDDCORE 109
VDDCR 12
VDDCR 52
VDDCR 76
VDD33 7
VDDIOB 37
OSCVDD12 4
VDDANA 16
VDDIO_J 29
VDDIO_J 45
VDDIO_J 60
VDDIO_J 74
VDDIO_J 94
VDDIO_S 65
VDDIO_S 97
VDDIO_S 117
OSCVSS 6
TESTMODE 81
EESDA/TMS 82
EESCL/TCK 83
IRQ/LATCH1 88
RUNLED/STATE_RUNLED/E2PSIZE 89
LINKACTLED1/TDI 90
LINKACTLED0/TDO 95
OSCO 1
OSCI 128
REG_EN 13
CLK_25 14
ERRLED/PME 17
WAIT_ACK/PME/LATCH0 18
RST# 21
NC 48
NC 75
NC 85
NC 86
SYNC0/LATCH0/PME 70
SYNC1/LATCH1/PME 39
GNDANA 15
GND 129
LAN9255/ZMX019
SB1
R1 10K
R2 10K
R3 10K
R4 10K
R5 10K
R6 33R
XTAL_OUT
XTAL_IN
ERRLED
RST
SYNC0
SYNC1
25MHz CRYSTAL
XTAL_IN X1 XTAL_OUT
C27 10pF
C28 10pF
RESET BUTTON
R9 10K RST
SW1
C31 0.1uF
INDICATION LEDS
PHY_IRQn R42 10K
EEPSIZE R7 330R LED1
ERRLED R8 1K LED2
R17 1K LED3
LED 3 is for CPU power indication
VDDTXRX2
VDDTXRX1
VCC_3V3
GND


ENC_SSC
ENC_CS
ENC_SCK
ENC_DATA
SW_INT
SWDCLK
SWDIO
SWO
RST
UART
UART_TX
UART_RX
To MCU
CLK_25
PHY_IRQn
Stepper
DIAG1
DIR
STEP
DIAG
ETH_SIG
TXA_N
TXA_P
RXA_N
RXA_P
TXB_N
TXB_P
RXB_P
RXB_N
I2C
SDA_
SCL_
U1C
A0/D15/AD15/DIGIO9/GPO9/MII_RXER
A1/ALELO/OE_EXT/MII_CLK25
A2/ALEHI/DIGIO10/GPO10/LINKACTLED2/LEDPOL2
A3/BE0/DIGIO11/GPI11/GPO11/MII_RXDV
A4/BE1/DIGIO12/GPI12/GPO12/MII_RXD0
CS/DIGIO13/GPI13/GPO13/MII_RXD1
WR/ENB/DIGIO14/GPI14/GPO14/MII_RXD2
RD/RD_WR/DIGIO15/GPI15/GPO15/MII_RXD3
TXNA
TXPA
RXNA
RXPA
RXPB
RXNB
TXPB
TXNB
D0/AD0/WD_STATE/SI/SIO0
D1/AD1/EOF/SO/SIO1
D2/AD2/SOF/SIO2
D3/AD3/WD_TRIG/SIO3
D4/AD4/DIGIO3/GPI3/GPO3/MII_LINK
D5/AD5/OUTVALID/SCS#
D6/AD6/DIGIO0/GPI0/GPO0/MII_RXCLK
D7/AD7/DIGIO1/GPI1/GPO1/MII_MDC
D8/AD8/DIGIO2/GPI2/GPO2/MII_MDIO
D9/AD9/LATCH_IN/SCK
D14/AD14/DIGIO8/GPI8/GPO8/MII_TXD3
D13/AD13/DIGIO7/GPI7/GPO7/MII_TXD2
D12/AD12/DIGIO6/GPI6/GPO6/MII_TXD1
D11/AD11/DIGIO5/GPI5/GPO5/MII_TXD0
D10/AD10/DIGIO4/GPI4/GPO4/MII_TXEN
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
QSPI_CS
QSPI_CLK
LAN9255/ZMX019
U1B
XIN32
XOUT32
UART_TX
UART_RX
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
SYNC0
SYNC1
DIAG1
SWDCLK
SWDIO
PA00
PA01
PA02
PA03
PA04
PA05
PA06
PA07
PA08
PA09
PA10
PA11
PA12
PA13
PA14
PA15
PA16
PA17
PA18
PA19
PA20
PA21
PA22
PA23
PA24
PA25
PA27
PA30
PA31
PB00
PB01
PB02
PB03
PB04
PB05
PB06
PB07
PB08
PB09
PB10
PB11
PB12
PB13
PB14
PB15
PB16
PB17
PB22
PB23
PB30
PB31
VSW
RESETN_E53
RBIAS
OPTO_IN
DIR
STEP
DIAG
R11
R12
33R QSPI_CLK
33R QSPI_CS
ENC_DATA
ENC_SCK
ENC_CS
ENC_DATA
CLK_25
PHY_IRQn
SWO
L6
VDD_CORE
R13
RST
12.1K
GND
LAN9255/ZMX019
VCC_3V3
R15
R16
10K
SDA_
SCL_
U2
VCC
WP
SDA
SCL
A0
A1
A2
VSS
R14
CAT24C256WI-GT3
EEPROM
32.768KHz CRYSTAL
X2
C32
C33
10pF
VCC_3V3
GND

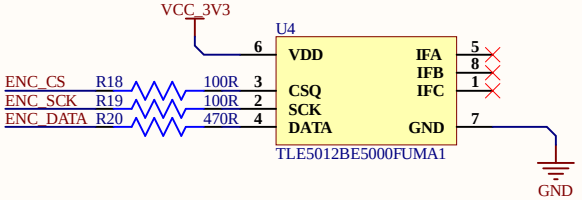
CONTROL SIGNAL



DECOUPLING CAPACITORS

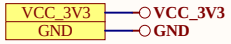


ENCODER FEEDBACK



Need to add Series Resistances

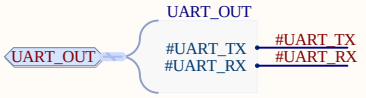
POWER SIGNALS



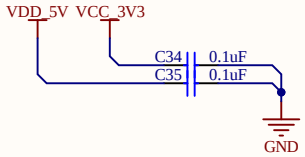
UART SIGNALS



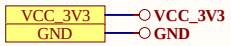
UART OUT SIGNALS



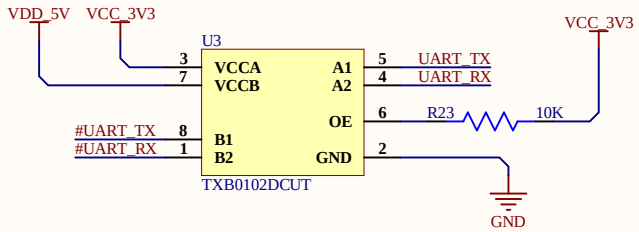
DECOUPLING CAPACITORS



POWER SIGNALS



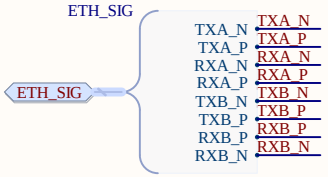
LEVEL SHIFTER



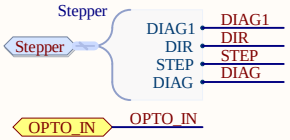
Setting the OE input high enables the device.
Datasheet: page-23

Will be changed with smaller part when library is available

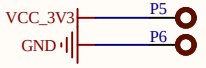
ETHERCAT SIGNALS



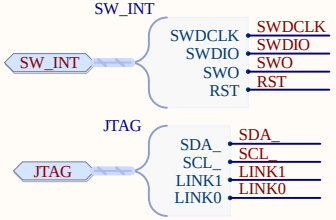
CONTROL SIGNALS



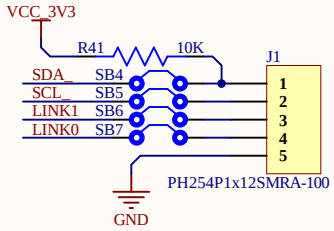
TEST POINTS



PROGRAMMING SIGNALS



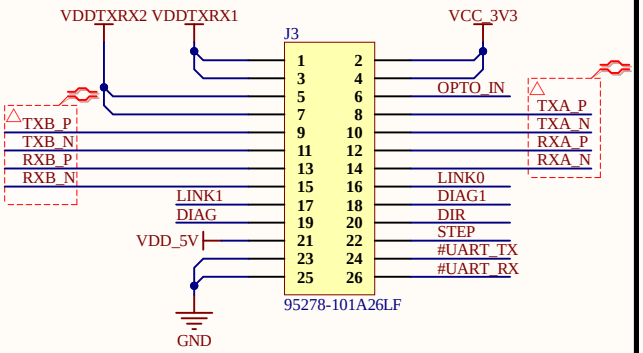
JTAG HEADER



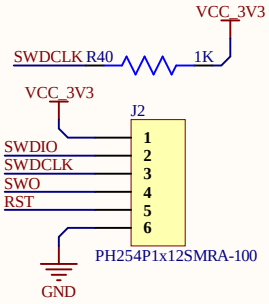
UART OUT SIGNALS



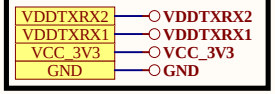
BOARD TO BOARD CONNECTOR

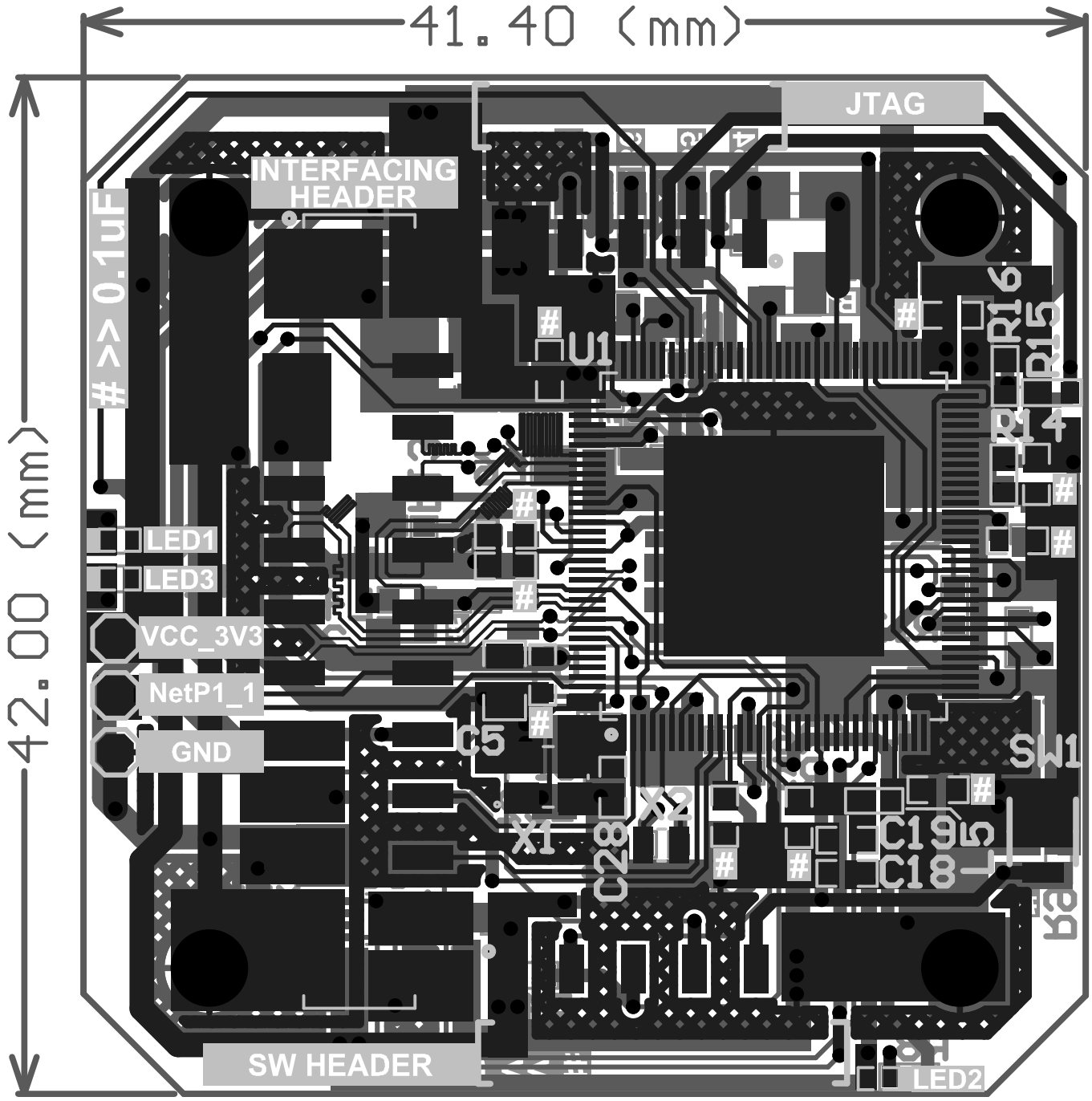


SW HEADER



POWER SIGNALS




41.40 (mm)
42.00 (mm)
JTAG
INTERFACING HEADER
>> 0.1uF
U1
R16
R15
R14
LED1
LED3
VCC_3V3
NetP1_1
GND
C5
X1
X2
C28
C19
C18
SW1
SW HEADER
LED2